ResuMetrics

Enterprise AI Resume Screening & Multi-Role ATS Platform

1. Executive Summary & Purpose

ResuMetrics is an enterprise-grade AI resume screening, talent analytics, and Applicant Tracking System (ATS). Built to streamline hiring workflows for recruiters, hiring managers, and candidates alike, ResuMetrics evaluates unstructured resumes (PDF and DOCX) against job requirements with sub-second latency. It delivers multi-dimensional hybrid AI scoring, timeline fraud auditing, role-specific portals, and cloud-ready hybrid persistence.

2. Dual-Role Experience & Core Workflows

- **Recruiter Workspace:** Batch drag-and-drop resume parsing, algorithmic weight tuning, interactive candidate leaderboard, radar comparison charts, automated shortlisting emails, and drag-and-drop Kanban ATS stages (Applied, Shortlisted, Interviewing, Rejected).
- **Applicant Portal:** Self-service ATS compatibility checker, single-resume gap analysis against targeted job descriptions, tailored missing skill roadmaps, and optimization tips.
- **1-Click Demo & Authentication:** Frictionless access via 1-click instant demo logins for both recruiter and applicant roles, supported by secure salted password hashing (pbkdf2:sha256) and session management.
- **Export & Reporting:** Instant generation of executive candidate summary PDFs via ReportLab and CSV spreadsheets compatible with Microsoft Excel and Google Sheets.

3. Hybrid AI Scoring Engine

The core scoring architecture blends three distinct, complementary ranking algorithms to eliminate keyword stuffing while preserving semantic comprehension:

- **TF-IDF Vectorization & Cosine Similarity (40% default):** Measures keyword frequency-inverse document frequency, highlighting rare domain-specific competencies over common vocabulary.
- **Exact Skill & N-Gram Extraction (35% default):** High-precision parser evaluating hard skills, specialized tools, and multi-word tech competencies extracted directly from job descriptions.
- **Dense Semantic Embeddings (25% default):** Neural sentence representations (SentenceTransformers / MiniLM) capturing conceptual similarity, recognizing equivalent roles and technologies without exact keyword matches.
- **Reciprocal Rank Fusion (RRF):** Mathematically fuses independent candidate rankings to ensure fair, unbiased, and normalized final scoring.

4. Advanced NLP & Integrity Auditing

- **Timeline Fraud Detection:** Audits claimed years of experience against detected employment date spans in job histories, flagging discrepancies and gaps.
- **Career Growth & Trajectory Tracking:** Detects rapid promotions, lateral leadership growth, and progressive job responsibility indicators.
- **Information Extraction (NER & Regex):** High-accuracy extraction of contact emails, phone numbers, academic degrees, certifications, and quantifiable metrics ($ saved, % growth).
- **Conversational RAG Chatbot:** Natural language query assistant allowing recruiters to ask questions across the parsed candidate pool in real time.

5. Infrastructure, Database & Cloud Deployment

- **Primary Database (MySQL / TiDB):** Persistent relational storage configured with account credentials (`root:nitesh123@localhost:3306/resumetrics`) storing candidate corpora, authentication tables, and screening history.
- **Self-Healing Storage Proxy:** Intelligent SQLite fallback proxy providing continuous zero-downtime execution in serverless cloud environments where local MySQL sockets are unreachable.
- **Serverless Vercel Routing:** WSGI ProxyFix and custom VercelRoutingMiddleware resolving clean URLs, rewrites, and serverless function entry points seamlessly.
- **Backend & Frontend Stack:** Python 3.12, Flask, Werkzeug, PyPDF, python-docx, ReportLab, Vanilla HTML5/CSS3, and Chart.js.